



Q/V Optical Encoder

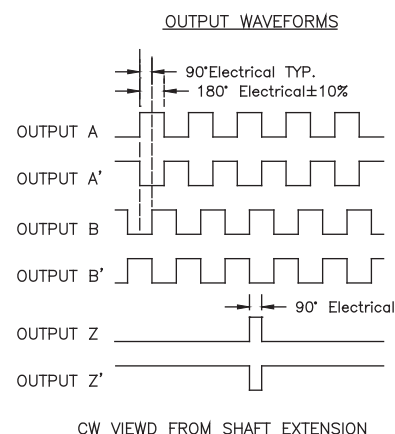
General information

- Output Channels – A,B, Z
- Differential Line Drivers
- IP 40 Rated
- Power Requirement -125Ma@5VDC
- 500kHz Maximum Operating Frequency
- Inertia – 0.0015 oz-in-sec²(0.01592Kg-cm²)
- Factory or user mountable

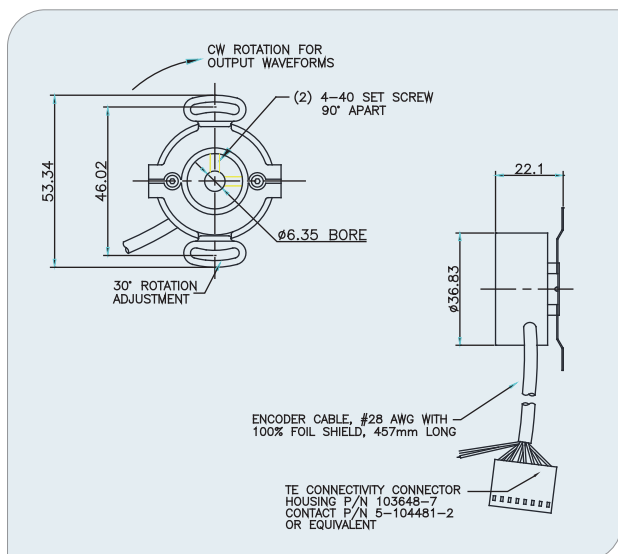
Encoder Connections

PIN	COLOR	FUNCTION
1	Red	VDC(+5V)
2	Black	Common
3	Brown	Output A
4	White	Output A'
5	Blue	Output B
6	Green	Output B'
7	Orange	Output Z
8	Yellow	Output Z'
N/C	BLK/WHT	Case Ground
N/C	Drain Wire	Cable Shield

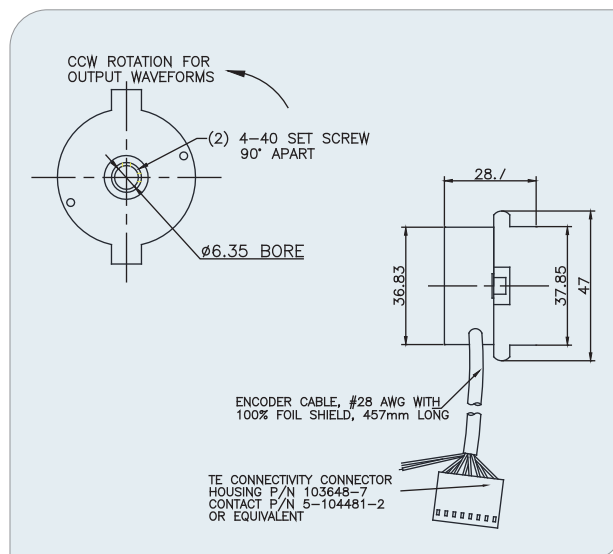
CATALOG NUMBER	STYLE	RESOLUTION
Q1	Standard	1000
V1	Small Dia.	1000
Q3	Standard	2000
V3	Small Dia.	2000
Q5	Standard	2500
V5	Small Dia.	2500
Q8	Standard	5000
V8	Small Dia.	5000



Mechanical outline –Q series



Mechanical outline –V series



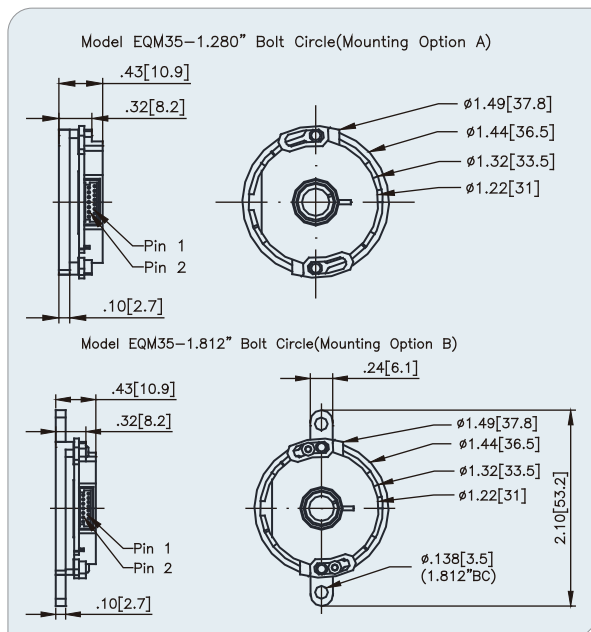
EQM35

Optical Encoder

General information

- Bearingless modular design
- Low profile assembled height
- Resolutions up to 8192 lines per revolution
- 4, 6, 8 or 10 pole commutation
- Easy Lock-n-Twist assembly feature
- Through shaft sizes up to 0.375" Diameter
- Up to 1Mhz Frequency Response
- High Noise Immunity
- RoHS Construction
- Hub to shaft uses two 3-48 set screws
- Hermitically sealed LED
- Multiple mounting options including resolver size 15

Mechanical



Ordering Information Sample: M-1000-4-A-A-L-A-A-A

Model Prefix	PPR	Poles	Electrical	Cover	Hub Size	Mounting	Index	Hardware
EQM	500	0	A=RS422	A=Hole in cover	C=5mm	A=1.280"BC	A=90°AB high	A=English
	512	4	B=RS422(ABZ)	B=No cover hole	D=6mm	B=1.812"BC		B=Metric
	1000	6	Open Collector(UVW)		E=8mm			C=No Hardware
	1024	8			L=0.25"			
	2000	10	C=Dual Voltage		N=0.375"			
	2048		RS422(ABZ)					
	2500		Open Collector(UVW)					
	4000							
	5000							
	5120							

Note:
 1.) 4 Poles has four states per revolution(2 pole pair), or two 360°electrical cycles per revolution.
 2.) RS422 is compatible with Renco options: TTL, PP, VC or LD. Open collector compatible with VO configurations.
 3.) 10 Pole only available for 500, 1000,2000,2500,5000 and 8000 PPR.
 4.) Consult factory for configurations not shown.



Electrical Specifications

Input Voltage	5 VDC ± 5%
Input Current Requirements	65mA Typ, 100mA Max plus Interface Loads
Input Ripple	2% Peak to Peak @ 5 VDC
Output Circuits	26C31 RS 422A Line Driver(TTL Compatible)Optional UVW Open Collector(No UVW)
Incremental Output Format	Quadrature A leading B for CCW rotation viewed from encoder top.Index Pulse true over A and B High
Max Operating Frequency	< 500ppr=500Khz or 15,000RPM 5000PPR-7999PPR=675Khz ≥ 8000PPR=1.0Mhz
Commutation Format	Three Phase 4,6,8 or 10POLES
Commutation Accuracy	± 2° mechanical
Interpolation Factors	1000/1024ppr=2x 2000/2048ppr=4x 2500PPR=5x 4000/4096ppr=8x 5000/5120ppr=10x 8000/8192ppr=16x

Environmental Specifications

Storage Temperature	-40 to 125 °C
Operation Temperature	-30 to 115 °C
IP Rating	40
Humidity	90% Non-Condensing
Vibration	20 g's @ 50 to 500CPS
Shock	50 g's @ 11ms Duration

Mechanical Specification

Hub Tolerance	0,+0.0006"
Recommended Shaft Tolerance	-0.0005,+0
Allowable Shaft Run Out	0.05mm,0.002" TIR
Allowable Axial Shaft Movement	± 0.25mm, ± 0.01"
Mounting	1.28", 1.812" Bolt Circle, Resolver mount
Dynamic Commutation Adjustment Range	30° mechanical
Moment of Inertia	8.0x10 ⁻⁶ oz-in-s ²